

Numerical PDEs Homework #0

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Problem 1: Finite differences

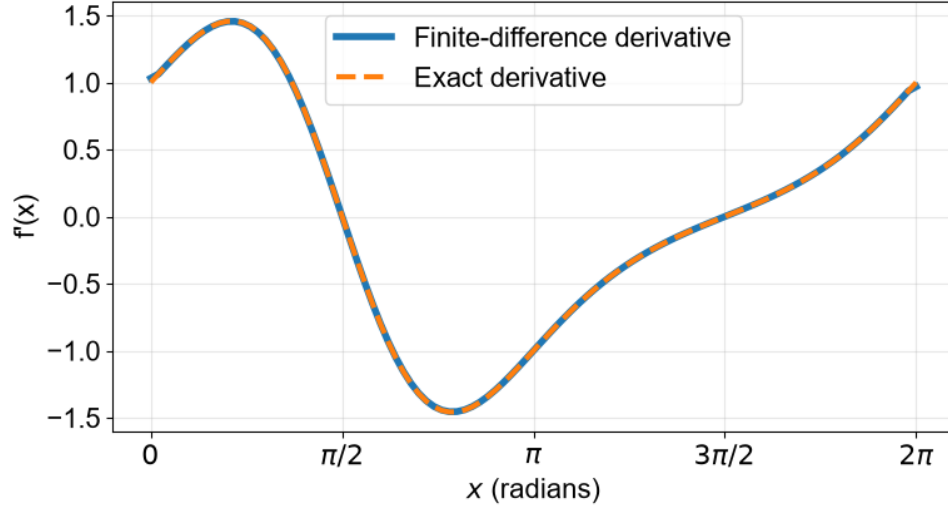


Figure 1: Finite difference approximation for $f(x) = \exp(\sin(x))$.

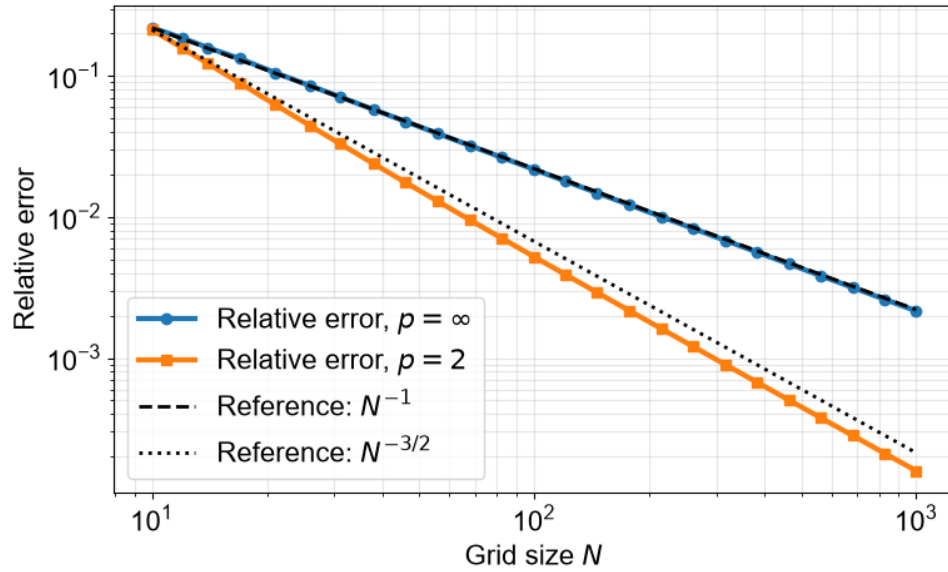


Figure 2: Relative-error convergence of first derivative finite differences for $f(x) = \exp(\sin(x))$

I implemented this solution using the exact centered difference equation for first derivatives with forward and backward differences on the endpoints (no matrix). The relative error slope of $3/2$

Problem 2: Second-order derivatives

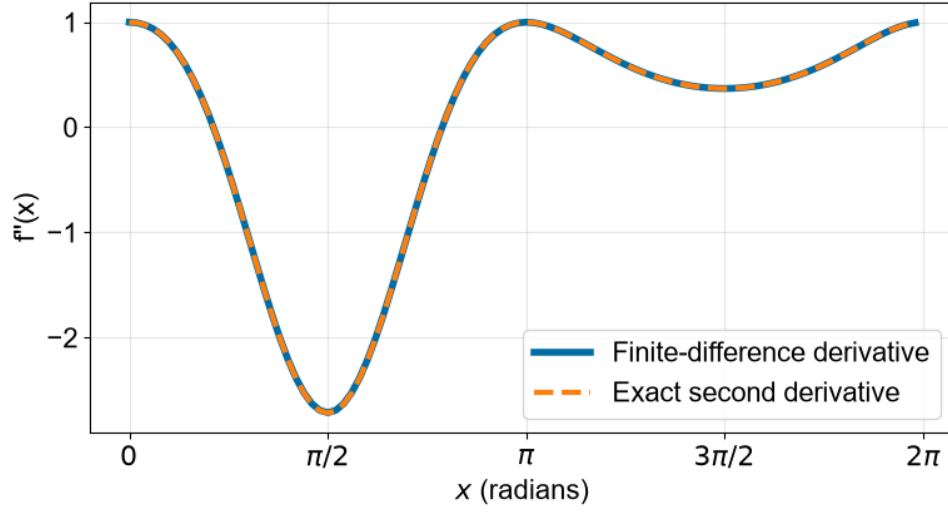


Figure 3: Numerical second derivative of $f(x) = \exp(\sin(x))$

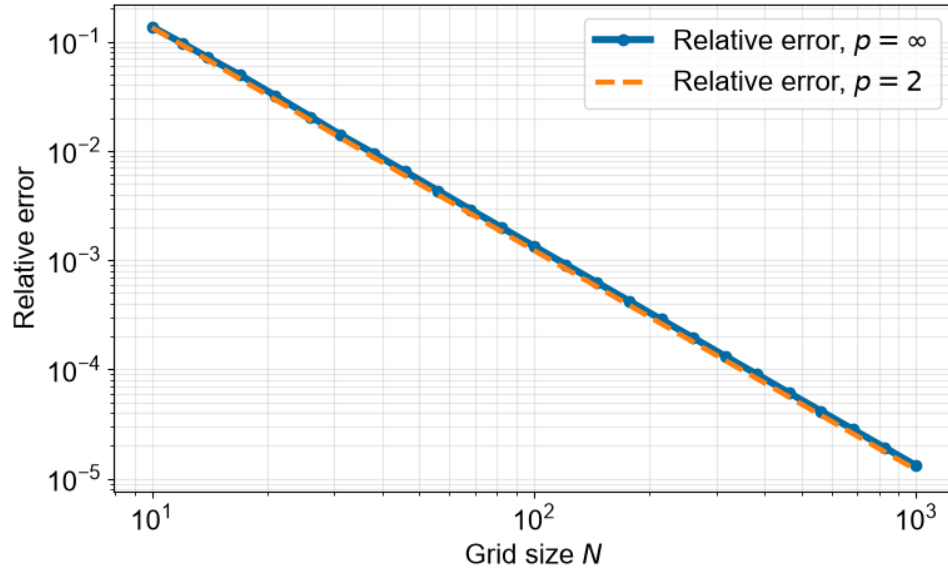


Figure 4: Relative-error convergence of second derivative finite difference for $f(x) = \exp(\sin(x))$.

I implemented this solution using the same method as part one with periodic assumptions instead of first-order differences at the endpoints.

Problem 3: ODE

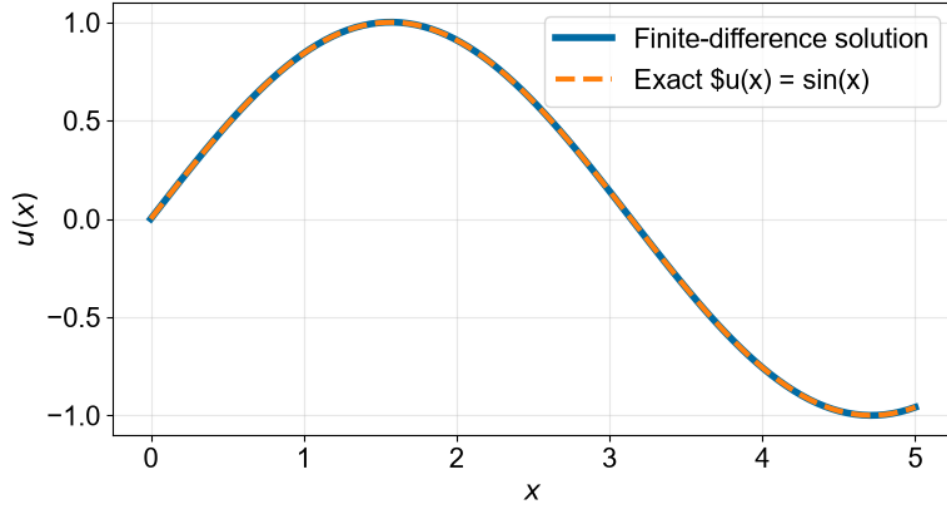


Figure 5: Finite-difference solution of $\frac{d^2u}{dx^2} + \sin(x)\frac{du}{dx} + u(x) = f(x)$, with $u(x) = \sin(x)$

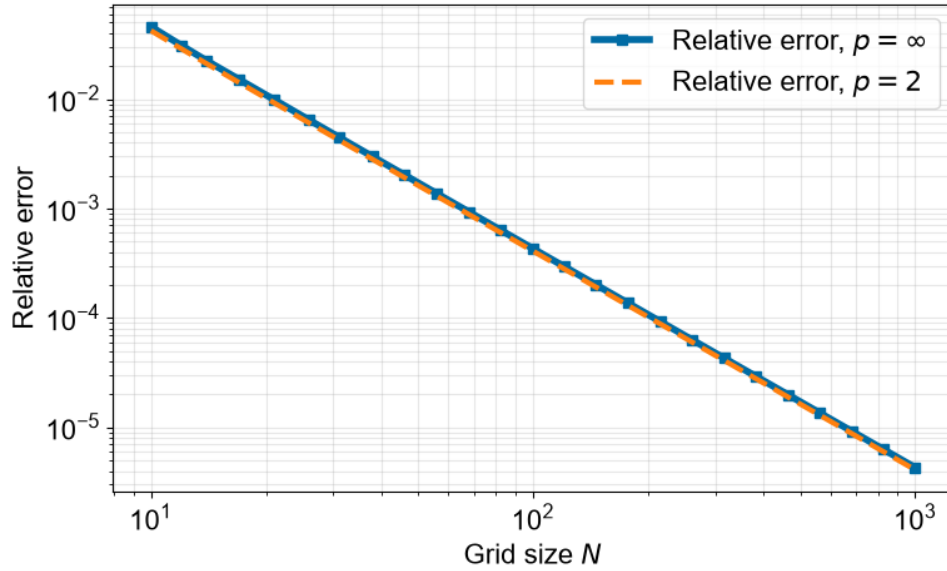


Figure 6: Relative-error convergence of ODE solution

I implemented this solution by constructing a sparse matrix and setting the boundary conditions on the right hand side. I used scipy for the sparse matrix and linear system solver.

Problem 4: Poisson Equation

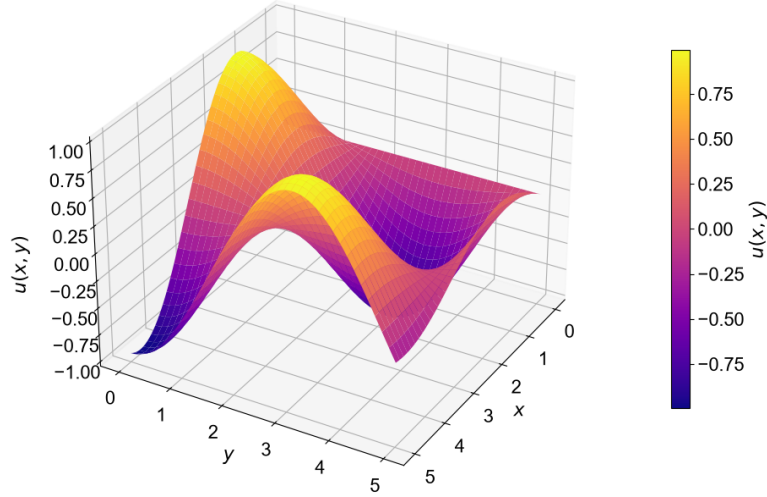


Figure 7: 3D finite-difference solution of $\nabla^2 u = f(x, y)$

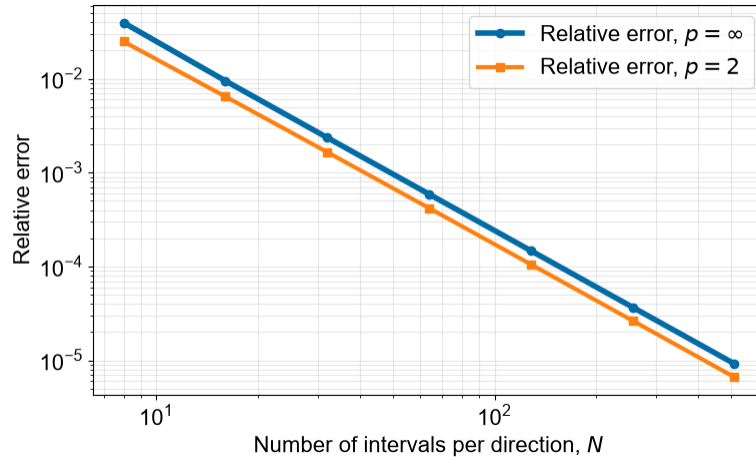


Figure 8: Relative-error convergence of the 2D Poisson solver

I implemented this solution by constructing a sparse matrix from kronecker products and flattening the unknowns and right hand side matrices. I set the boundary conditions in the right hand side using a boundary index mapping.

Neumann boundary conditions do not work because they do not give a unique solution. Since they only specify derivatives, for any solution u , $u + C$ is also a solution. In linear algebra, the Neumann conditions gives a laplacian that is not invertible and therefore cannot satisfy the linear system with a unique solution. A solution to this problem could be setting a single condition for $u(x, y)$ to effectively fix a single unique solution in place.